

ASSIGNMENTS 2 AND 3 – DISCRETE MATHEMATICS (MAT 2138)

1. Let A and B be two sets, and $A \times B$ their Cartesian product. Show that there exist functions $\pi_A: A \times B \rightarrow A$ and $\pi_B: A \times B \rightarrow B$ satisfying the following: If C is any set and $f: C \rightarrow A$ and $g: C \rightarrow B$ are any two functions, then there exists a function $h: C \rightarrow A \times B$ such that $f = \pi_A \circ h$ and $g = \pi_B \circ h$. (3M)
2. A full binary tree is a binary tree in which every vertex has either 0 or 2 children. Let C_n be the number of full binary trees with $n + 1$ leaves. Obtain a recurrence relation for C_n , and hence derive the ordinary generation function $C(x)$ for C_n . Using $C(x)$, find a closed-form expression for C_n . (3M)
3. Let G and H be two graphs on the same vertex set $V = \{v_1, \dots, v_n\}$. Let A and B be the adjacency matrices of G and H , respectively. Show that AB is the adjacency matrix of a simple undirected graph if and only if the following hold: (4M)
 - (i) G and H have no common edges – i.e., if two vertices are adjacent in G , then they are non-adjacent in H .
 - (ii) If there exists k such that $v_i \sim_G v_k$ and $v_k \sim_H v_j$, then there exists l such that $v_i \sim_H v_l$ and $v_l \sim_G v_k$.
 - (iii) There exists at most one k such that $v_i \sim_G v_k$ and $v_k \sim_H v_j$.